

DOCUMENTATION

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Conditional Logic (if / elif / else)

Used to make decisions in your code.

```
age = 18
```

```
if age < 13:
    print("Child")
elif age < 18:
    print("Teenager")
else:
    print("Adult")
```

How it works

- if → first condition checked
- elif → additional conditions
- else → runs if none are true

Comparison Operators

Used to compare values.

Operator	Meaning	Example
==	Equal to	5 == 5 → True
!=	Not equal to	5 != 3 → True

Operator	Meaning	Example
>	Greater than	7 > 2
<	Less than	2 < 7
>=	Greater or equal	5 >= 5
<=	Less or equal	4 <= 6

Example:

```
x = 10
```

```
y = 5
```

```
print(x > y) # True
```

```
print(x == y) # False
```

Lists

Lists store multiple values in one variable.

```
fruits = ["apple", "banana", "orange"]
```

Access items

```
print(fruits[0]) # apple
```

Add items

```
fruits.append("grape")
```

Remove items

```
fruits.remove("banana")
```

Loop through a list

```
for fruit in fruits:  
    print(fruit)
```

for Loops

Used to repeat actions for each item in a sequence.

```
numbers = [1, 2, 3, 4]
```

```
for number in numbers:  
    print(number)
```

Using range()

```
for i in range(5):  
    print(i)
```

Output:

```
0  
1  
2  
3  
4
```

while Loops

Repeats while a condition is True.

```
count = 1
```

```
while count <= 5:  
    print(count)  
    count += 1
```

Important

Make sure the condition eventually becomes False, or the loop will run forever.

Basic Error Handling (try-except)

Prevents your program from crashing when errors happen.

```
try:  
    number = int(input("Enter a number: "))  
    print(number)  
except ValueError:  
    print("That's not a valid number.")
```

Common flow

- try → code that might fail
- except → what to do if an error occurs

Combined Example

```
numbers = [10, 20, 0, 5]
```

```
for num in numbers:
```

```
    try:
```

```
        result = 100 / num
```

```
        if result > 10:
```

```
            print("Large result:", result)
```

```
        else:
```

```
            print("Small result:", result)
```

```
    except ZeroDivisionError:
```

```
        print("Cannot divide by zero.")
```

This example uses:

- lists
- for loops
- conditional logic
- comparison operators
- error handling

1. Number Guessing Game with Hints

```
import random
```

```
secret_number = random.randint(1, 100)
```

```
print("Guess the number between 1 and 100!")
```

```
while True:
```

```
    try:
```

```
        guess = int(input("Enter your guess: "))
```

```
        if guess < secret_number:
```

```
            print("Too low!")
```

```
        elif guess > secret_number:
```

```
        print("Too high!")
    else:
        print("Correct! You guessed the number.")
        break

except ValueError:
    print("Please enter a valid number.")
```

2. Age Categorizer

```
try:
    age = int(input("Enter your age: "))

    if age < 0:
        print("Invalid age")
    elif age <= 12:
        print("Child")
    elif age <= 19:
        print("Teenager")
    elif age <= 64:
        print("Adult")
    else:
        print("Senior")

except ValueError:
    print("Please enter a valid age.")
```

3. Shopping List Manager

```
shopping_list = []

while True:
    print("\n1. Add item")
    print("2. Remove item")
    print("3. View list")
    print("4. Exit")

    choice = input("Choose an option: ")

    if choice == "1":
```

```
    item = input("Enter item to add: ")
    shopping_list.append(item)
    print(f"{item} added.")

elif choice == "2":
    item = input("Enter item to remove: ")

    if item in shopping_list:
        shopping_list.remove(item)
        print(f"{item} removed.")
    else:
        print("Item not found.")

elif choice == "3":
    print("Shopping List:")
    for item in shopping_list:
        print("-", item)

elif choice == "4":
    print("Goodbye!")
    break

else:
    print("Invalid option.")
```

4. Multiplication Table Generator

```
try:
    number = int(input("Enter a number: "))

    for i in range(1, 11):
        print(f"{number} x {i} = {number * i}")

except ValueError:
    print("Please enter a valid number.")
```

5. Simple Login System

```
correct_username = "admin"
correct_password = "python123"
```

```
username = input("Username: ")
password = input("Password: ")

if username == correct_username and password == correct_password:
    print("Login successful!")
else:
    print("Invalid username or password.")
```

6. List Operations Practice

```
numbers = [1, 2, 3, 4, 5]
```

```
# Add item
```

```
numbers.append(6)
```

```
# Insert item
```

```
numbers.insert(0, 0)
```

```
# Remove item
```

```
numbers.remove(3)
```

```
# Sort list
```

```
numbers.sort()
```

```
# Reverse list
```

```
numbers.reverse()
```

```
print(numbers)
```

```
# Length of list
```

```
print("Length:", len(numbers))
```

```
# Check if item exists
```

```
print(4 in numbers)
```

7. Error Handling for User Inputs

```
while True:
```

```
    try:
```

```
number = float(input("Enter a number: "))  
print("You entered:", number)  
break
```

```
except ValueError:  
    print("Invalid input. Please enter a numeric value.")
```

Bonus Challenge

Combine several concepts into one mini project:

- Ask the user to log in
- Let them manage a shopping list
- Handle invalid inputs
- Use loops and conditions throughout

This is excellent practice for beginner Python programming.

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